

Quiz (Habanero)

- Q1.** Factor the trinomial $x^2 + 5x + 6$.
(A) $(x + 3)(x + 2)$
(B) $x(x + 5) + 6$
(C) $x^2 + (5 + x)$
(D) $x(x + 6) + 5$
(E) $x^2 + x + 5$
- Q2.** What is the factored form of $2x^2 + 7x + 3$?
(A) $(2x + 1)(x + 3)$
(B) $x(2x + 7) + 3$
(C) $(x + 1)(2x + 3)$
(D) $2(x^2 + 3x) + 7x$
(E) $2x(x + 3) + 1$
- Q3.** A rain gauge is used to measure rainfall. If x is the amount of rainfall, $x^2 + 4x + 4$ represents the amount of water collected. Factor this expression.
(A) $x(x + 2) + 4$
(B) $x^2 + (4 + x)$
(C) $(x + 2)^2$
(D) $2x(x + 2)$
(E) $x^2 + x + 4$
- Q4.** A temperature change of $x^2 - 7x + 12$ degrees is observed. Factor this expression.
(A) $(x - 3)(x - 4)$
(B) $x(x - 7) + 12$
(C) $x^2 - (7 + x)$
(D) $x(x - 3) - 4$
(E) $x^2 - x - 12$
- Q5.** If $3x^2 + 2x - 5$ represents the rate of change of a weather system, what is the factored form?
(A) $(3x + 5)(x - 1)$
(B) $(3x - 1)(x + 5)$
(C) $3(x^2 + 2x) - 5$
(D) $3x(x + 2) - 5$
(E) $3x^2 + (2 + 3x)$
- Q6.** A group of scientists are studying an ecosystem. They model the population using $2x^2 + x - 1$. Factor this trinomial.
(A) $(2x - 1)(x + 1)$
(B) $2(x^2 + 3x) - 5x$
(C) $x(2x + 1) - 1$
(D) $2x(x + 1) - 3$
(E) $x^2 + (1 + 2x)$
- Q7.** What is the factored form of $x^2 - 9x + 20$?
(A) $(x - 4)(x - 5)$
(B) $x(x - 9) + 20$
(C) $x^2 - (9 + x)$
(D) $x(x - 4) - 5$
(E) $(x - 5)(x - 4)$
- Q8.** The expression $x^2 + 2x - 6$ represents the amount of rainfall in an area. Factor this expression.
(A) $x(x + 2) - 6$
(B) $x^2 + (2 + x)$
(C) $(x + 3)(x - 2)$
(D) $x(x + 6) - 2$
(E) $2x(x + 1) - 3$

Open Ended Questions (FRQ)

- Q9.** Factor the trinomial $x^2 - 4x - 3$. Provide all steps and the final factored form.
- Q10.** The cost of water treatment can be modeled using $2x^2 + 5x - 3$, where x is the amount of chemicals used. Factor this expression and provide a brief explanation of your steps.

Chef's Tips

- Tip 1: Factor out greatest common factors before applying other factoring methods
- Tip 2: Check for two binomials that when multiplied result in the given trinomial
- Tip 3: Use the quadratic formula if necessary